

Unit-3: Knowledge Representation

Table of Topics and Subtopics

Main Topic	Subtopics	Key Ideas to Understand
3.1 Representations and Mappings	Knowledge representation concept, Mapping real world knowledge into AI models	How real-world knowledge is converted into machine-understandable form
3.2 Approaches to Knowledge Representation	Simple relational knowledge, Inheritable knowledge, Inferential knowledge, Procedural knowledge	Different ways knowledge can be structured in AI
3.3 Representing Simple Facts in Logic	Propositional logic, logical statements, truth values	Representing basic facts using logic
3.4 Representing Instance and ISA Relationships	Instance relationship, ISA hierarchy, inheritance	Representing class and object relationships
3.5 Computable Functions and Predicates	Functions in logic, predicates, logical expressions	Representing relationships between objects
3.6 Resolution	Resolution rule, logical inference, theorem proving	Logical reasoning technique used in AI
3.7 Procedural vs Declarative Knowledge	Procedural knowledge, declarative knowledge, differences	Two major knowledge types used in AI
3.8 Logic Programming	Concept of logic programming, rule-based programs	Programming based on logical rules
3.9 Forward Reasoning	Forward chaining inference	Data-driven reasoning
3.10 Backward Reasoning	Backward chaining inference	Goal-driven reasoning

3.1 Representations and Mappings

Knowledge Representation (KR) is a fundamental concept in Artificial Intelligence. It focuses on **how knowledge about the real world is represented inside a computer so that the AI system can reason, learn, and make decisions.**

This topic explains:

- what knowledge representation is
- how real-world information is converted into AI models
- how AI systems interpret and use that knowledge

1. Knowledge Representation Concept

Meaning : Knowledge Representation (KR) is the method of **storing and organizing knowledge in a form that a computer system can understand and use for reasoning and decision making.**

In AI, knowledge is not just stored as raw data; it must be **structured in a meaningful way** so that the system can:

- interpret information
- draw conclusions
- solve problems

Simple Definition : Knowledge representation is the process of **encoding knowledge about the world into a computer system.**

Example

Human knowledge:

"All humans are mortal."

In AI representation:

Human(x) → Mortal(x)

This logical representation allows the AI system to **perform reasoning**.

Why Knowledge Representation is Important

AI systems must perform tasks that require intelligence.

For this, they need knowledge about:

- objects
- relationships
- rules
- events

Without knowledge representation, AI systems cannot reason or make decisions.

Functions of Knowledge Representation

Function	Description
Knowledge storage	storing information about the world
Reasoning support	enabling logical inference
Problem solving	helping AI systems solve problems
Decision making	choosing correct actions

2. Types of Knowledge Represented in AI

Different kinds of knowledge can be represented in AI systems.

Type	Description	Example
Declarative knowledge	facts and information	Paris is capital of France
Procedural knowledge	how to perform tasks	steps to cook rice
Heuristic knowledge	experience-based rules	shortcuts used in chess
Meta knowledge	knowledge about knowledge	reasoning strategies

3. Mapping Real-World Knowledge into AI Models

Meaning: Mapping is the process of **converting real-world knowledge into a form that computers can process**.

In other words, mapping translates **human knowledge into machine-readable representation**.

Mapping Process

The mapping process involves several steps.

Step	Description
Observation	collect real-world information
Concept identification	identify objects and relationships
Representation	encode knowledge using formal structures
Implementation	store representation in AI system

Example

Real-world statement:

"Dogs are animals."

AI representation:

Dog → is-a → Animal

This mapping converts natural language knowledge into a **structured representation**.

Example: Medical Knowledge Mapping

Real-world knowledge:

If a patient has high temperature and cough, they may have an infection.

AI representation:

IF temperature > 38 AND cough

THEN infection

The rule allows the AI system to **reason about medical diagnosis**.

4. Mapping Real-World Knowledge into AI Structures

Several structures are used to represent mapped knowledge.

Structure	Description
Logical representation	knowledge expressed using logic
Semantic networks	graphical representation of relationships
Frames	structured knowledge objects
Production rules	IF-THEN rule representation

Example: Semantic Network

Real-world knowledge:

A dog is an animal.

Representation:

Dog → is-a → Animal

Animal → has → Life

Nodes represent objects, edges represent relationships.

Example: Frame Representation

Frame for a **car**:

Attribute	Value
Type	Vehicle
Wheels	4
Fuel	Petrol

Frames store knowledge as **structured objects with attributes**.

5. Mapping AI Knowledge Back to Real World

The mapping process also works in the opposite direction.

AI systems interpret stored knowledge and **convert it into real-world actions or decisions**.

This is called **reverse mapping**.

Example

AI system knowledge:

IF patient has fever AND cough

THEN infection

Real-world output:

Doctor should check for infection.

The AI converts internal representation into **useful real-world advice**.

6. Challenges in Knowledge Representation

Representing knowledge in AI is not always easy.

Several challenges exist.

Challenge	Explanation
Knowledge complexity	real-world knowledge is very complex
Uncertainty	information may be incomplete
Dynamic knowledge	knowledge may change over time
Representation efficiency	representation must support reasoning

7. Characteristics of Good Knowledge Representation

A good knowledge representation system should satisfy several properties.

Property	Description
Representational adequacy	ability to represent required knowledge
Inferential adequacy	ability to derive new knowledge
Inferential efficiency	ability to reason efficiently
Acquisitional efficiency	ability to add new knowledge easily

8. Example: AI Knowledge Representation in Self-Driving Cars

Real-world knowledge:

- traffic signals
- road rules
- object detection

AI representation:

Knowledge	Representation
traffic light red	stop vehicle
pedestrian detected	slow down
obstacle ahead	change lane

This mapping allows the AI system to make **intelligent driving decisions**.

Summary

Concept	Meaning
Knowledge Representation	storing knowledge in AI systems
Mapping	converting real-world knowledge to AI models
Representation structures	logic, semantic networks, frames, rules
Reverse mapping	AI knowledge converted to real-world decisions

Knowledge Representation is essential because it **allows AI systems to understand, reason about, and interact with the real world**.

3.2 Approaches to Knowledge Representation

In Artificial Intelligence, **knowledge can be represented in different ways depending on the type of information and reasoning required**. These methods are called **approaches to knowledge representation**.

Each approach organizes knowledge differently so that AI systems can **store information, understand relationships, and perform reasoning**.

The main approaches are:

1. Simple Relational Knowledge
2. Inheritable Knowledge
3. Inferential Knowledge
4. Procedural Knowledge

Overview of Knowledge Representation Approaches

Approach	Main Idea	Representation Form
Simple Relational Knowledge	knowledge stored as relations between objects	tables or database structures
Inheritable Knowledge	hierarchical relationships between classes and objects	semantic networks
Inferential Knowledge	knowledge represented as logical rules	predicate logic
Procedural Knowledge	knowledge stored as procedures or instructions	programs or algorithms

1. Simple Relational Knowledge

Meaning : Simple relational knowledge represents knowledge as **relationships between objects using tables or relational structures**.

It is the simplest form of knowledge representation.

Information is stored as **relations (similar to database tables)** where each row represents a fact.

Structure

Knowledge is organized using **attributes and values**.

Example structure:

Object	Attribute	Value
Dog	Type	Animal
Dog	Legs	4
Dog	Sound	Bark

This representation describes relationships between objects and their properties.

Example

Consider knowledge about students.

Student	Course	Grade
Alice	AI	A
Bob	AI	B
Charlie	AI	A

This table represents relationships between students, courses, and grades.

Advantages

Advantage	Explanation
Easy to understand	similar to database tables
Simple implementation	easy to store and retrieve data
Suitable for structured data	works well with fixed attributes

Limitations

Limitation	Explanation
Limited reasoning capability	difficult to derive new knowledge
Poor representation of complex relationships	cannot easily represent hierarchical knowledge

2. Inheritable Knowledge

Meaning : Inheritable knowledge represents knowledge in a **hierarchical structure where properties are inherited from parent classes to child classes**.

This approach organizes knowledge using **class-subclass relationships**.

Objects inherit characteristics from more general categories.

Representation : This type of knowledge is often represented using **semantic networks or hierarchical structures**.

Example

Consider the following hierarchy.

```
Animal
|
|-- Bird
|
|-- Sparrow
```

Properties:

Class	Property
Animal	can breathe
Bird	has wings
Sparrow	can fly

Since Sparrow is a Bird and Bird is an Animal, Sparrow inherits properties from both.

Inheritance Concept

Inheritance allows properties to **propagate down the hierarchy**.

Example:

Level	Property
Animal	living organism
Bird	has feathers
Sparrow	small bird

Sparrow automatically inherits:

- living organism
- has feathers

Example Representation

Sparrow → is-a → Bird

Bird → is-a → Animal

Properties flow downward through the hierarchy.

Advantages

Advantage	Explanation
Efficient representation	avoids repetition of properties
Natural representation	similar to human knowledge organization
Supports inheritance	allows property sharing

Limitations

Limitation	Explanation
exceptions difficult to represent	e.g., penguins cannot fly
complex hierarchies may be difficult to manage	

3. Inferential Knowledge

Meaning : Inferential knowledge represents knowledge in a way that allows the system to **derive new facts using logical inference**.

Instead of storing all facts explicitly, rules are used to **generate new knowledge**.
This approach relies on **logic-based representation**.

Representation

Knowledge is expressed using **logical rules and predicates**.

Example format: IF condition THEN conclusion

or logical form: $P \rightarrow Q$

Example

Knowledge:

All humans are mortal

Socrates is a human

Logical representation:

$\text{Human}(x) \rightarrow \text{Mortal}(x)$

$\text{Human}(\text{Socrates})$

Inference: $\text{Mortal}(\text{Socrates})$

The system derives new knowledge through reasoning.

Inference Mechanism

AI systems use **inference engines** to apply logical rules.

Two main inference methods:

Method	Description
Forward chaining	start with facts and derive conclusions
Backward chaining	start with goal and verify facts

Advantages

Advantage	Explanation
supports logical reasoning	derives new knowledge
reduces need to store all facts	knowledge can be inferred
powerful for expert systems	widely used in rule-based systems

Limitations

Limitation	Explanation
reasoning can be computationally expensive	
complex rule systems may become difficult to manage	

4. Procedural Knowledge

Meaning : Procedural knowledge represents knowledge as **procedures or sequences of actions that describe how to perform tasks**.

It focuses on **how something is done rather than what is known**.

This knowledge is often implemented as **programs or algorithms**.

Representation

Procedural knowledge is stored in:

- functions
- programs
- procedures
- scripts

Example

Procedure for making tea:

1. Boil water
2. Add tea leaves
3. Add sugar
4. Add milk

This describes **how to perform the task**.

Example in Programming

A function to calculate factorial:

```
function factorial(n):
```

```
    if n == 1 return 1
```

```
    else return n × factorial(n-1)
```

The knowledge is represented as a **procedure**.

Advantages

Advantage	Explanation
efficient execution	direct instructions
easy to implement in programs	natural for programming
good for task-oriented knowledge	

Limitations

Limitation	Explanation
difficult to modify	changing procedures may affect system behavior
knowledge not explicitly visible	harder to interpret

Comparison of Knowledge Representation Approaches

Approach	Focus	Example
Simple Relational	facts stored as relations	database tables
Inheritable	hierarchical knowledge	class-subclass hierarchy
Inferential	reasoning using logic	rule-based systems
Procedural	how to perform tasks	algorithms and programs

Real-World Example Combining These Approaches

AI system for **medical diagnosis**:

Approach	Use
Relational knowledge	patient records
Inheritable knowledge	disease classification
Inferential knowledge	diagnosis rules
Procedural knowledge	treatment procedures

Combining these approaches allows AI systems to **represent knowledge effectively and perform intelligent reasoning**.

Summary

Concept	Meaning
Simple Relational Knowledge	facts stored in relational structures
Inheritable Knowledge	hierarchical property inheritance
Inferential Knowledge	logical rules used for reasoning

Procedural Knowledge	knowledge stored as procedures
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These approaches provide **different ways to structure knowledge in AI systems**, enabling them to store information and reason about it.

3.3 Representing Simple Facts in Logic

This topic introduces:

- propositional logic
- logical statements
- truth values
- logical operators.

Representing Simple Facts in Logic : In Artificial Intelligence, **logic is used to represent facts and relationships in a precise and formal way** so that machines can reason about them.

When knowledge is represented using logical expressions, an AI system can:

- analyze facts
- draw conclusions
- perform reasoning

The simplest logical representation used in AI is **Propositional Logic**.

Overview of Logical Representation

Concept	Meaning
Logical Statement	a sentence that is either true or false
Truth Value	determines whether a statement is true or false
Propositional Logic	logic that deals with simple statements
Logical Operators	symbols used to combine statements

1. Logical Statements

Meaning : A **logical statement (proposition)** is a sentence that has a **definite truth value**.

It must be either:

- **True**
- **False**

but not both.

Examples of Logical Statements

Statement	Truth Value
The Earth revolves around the Sun	True
5 + 3 = 9	False
Paris is the capital of France	True

These statements are propositions because they can be evaluated as **true or false**.

Non-Logical Statements

Some sentences cannot be considered logical statements.

Examples:

Sentence	Reason
Close the door	command
How are you?	question
x + y = 10	unknown variables

These cannot be assigned definite truth values.

2. Propositional Logic

Meaning : Propositional Logic is a branch of logic used to represent knowledge using **simple propositions and logical operators**.

Each proposition represents a **basic fact about the world**.

Representation

Propositions are usually represented using symbols.

Example:

Symbol	Meaning
P	It is raining
Q	The road is wet

Logical reasoning can be performed using these symbols.

Example

Knowledge:

1. If it rains, the road becomes wet.
2. It is raining.

Representation:

$P \rightarrow Q$

P

Conclusion: Q

This means the road is wet.

3. Truth Values

Meaning : A **truth value** indicates whether a proposition is **true or false**.

Propositional logic uses two truth values:

Symbol	Meaning
T	True
F	False

Example

Statement: P: "It is raining"

Possible truth values:

Situation	Truth Value
Rain occurring	True
No rain	False

Truth Table Concept

Truth tables show the truth value of logical expressions.

Example for **NOT operator**.

P	$\neg P$
True	False
False	True

4. Logical Operators: Logical operators are used to **combine propositions to form complex expressions**.

The most common logical operators are:

Operator	Symbol	Meaning
----------	--------	---------

NOT	$\neg$	negation
AND	$\wedge$	conjunction
OR	$\vee$	disjunction
IMPLIES	$\rightarrow$	implication
IFF	$\leftrightarrow$	biconditional

1. NOT Operator

Negation reverses the truth value.

Example:

Statement	Meaning
P	It is raining
$\neg P$	It is not raining

Truth table:

P	$\neg P$
T	F
F	T

2. AND Operator : The AND ($\wedge$) operator is true only if **both statements are true**.

Example:

P: It is raining

Q: The road is wet

Expression: $P \wedge Q$

Truth table:

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

3. OR Operator : The OR ($\vee$) operator is true if **at least one statement is true**.

Example:

P: It is raining

Q: It is cloudy

Expression: $P \vee Q$

Truth table:

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

4. Implication ($\rightarrow$)

Implication represents **cause-effect relationships**.

Example: $P \rightarrow Q$

Meaning: "If P is true, then Q must also be true."

Example: If it rains, the ground becomes wet

Truth table:

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

5. Biconditional ($\leftrightarrow$)

This operator means **both statements are equivalent**.

Example: $P \leftrightarrow Q$

Meaning: P is true **if and only if** Q is true.

Truth table:

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

Representing Basic Facts Using Logic

AI systems convert real-world knowledge into logical expressions.

Example 1: Weather Knowledge

Fact: If it rains, the ground becomes wet.

Logical representation: $\text{Rain} \rightarrow \text{WetGround}$

Example 2: Student Knowledge

Facts:

- All students study.
- John is a student.

Representation:

$\text{Student}(\text{John})$

$\text{Student}(x) \rightarrow \text{Studies}(x)$

Inference: $\text{Studies}(\text{John})$

Example 3: Traffic Rule

Fact: If the traffic light is red, vehicles must stop.

Logical form: $\text{RedLight} \rightarrow \text{StopVehicle}$

Importance of Logical Representation in AI

Logical representation allows AI systems to:

- perform reasoning

- derive conclusions
- solve problems logically

This approach is used in:

AI Application	Example
Expert systems	medical diagnosis
Automated reasoning	theorem proving
Knowledge-based systems	rule-based AI

Limitations of Propositional Logic

Limitation	Explanation
Cannot represent complex relationships	limited to simple statements
Cannot express quantifiers	like "all", "some"
Large knowledge bases become difficult to manage	

Because of these limitations, AI uses **Predicate Logic**, which is more powerful.

Summary

Concept	Meaning
Logical Statement	sentence with truth value
Truth Value	true or false evaluation
Propositional Logic	logic using simple statements
Logical Operators	AND, OR, NOT, implication

Representing facts using logic enables AI systems to **store knowledge in a structured form and perform reasoning to derive new information.**

3.4 Representing Instance and ISA Relationships

This topic explains:

- class-object relationships
- inheritance
- semantic hierarchies.

3.4 Representing Instance and ISA Relationships: Instance relationship, ISA hierarchy (in depth), inheritance, Representing relationships between objects

Representing Instance and ISA Relationships : In Artificial Intelligence, knowledge is often organized using **relationships between objects and categories**. Two important relationships used in knowledge representation are:

- **Instance relationship**
- **ISA relationship**

These relationships help AI systems understand **how objects belong to categories and how properties are inherited.**

They are commonly represented using **semantic networks and hierarchical structures.**

Overview

Concept	Meaning
Instance Relationship	relationship between an object and its class
ISA Relationship	relationship between a subclass and a superclass
ISA Hierarchy	hierarchical structure of classes
Inheritance	transfer of properties from parent class to child class

1. Instance Relationship

Meaning : An **instance relationship** represents the relationship between a **specific object (instance)** and the **class it belongs to**.

It shows that an object is a **member of a particular category**.

The relationship is written as: instance-of

Example

Consider the object **"Tweety"**.

Tweety instance-of Bird

This means: Tweety is an individual object that belongs to the class **Bird**.

Representation

Tweety → instance-of → Bird

More Examples

Object	Instance-of	Class
Tweety	instance-of	Bird
Tom	instance-of	Cat
Rover	instance-of	Dog

Each object belongs to a specific class.

Real-World Example

Student system:

Rahul instance-of Student

Here:

- Rahul is a specific person
- Student is the class

2. ISA Relationship

Meaning : The **ISA relationship** represents a relationship between a **subclass and a superclass**.

It indicates that one class **is a type of another class**.

Example: Bird ISA Animal

Meaning: A bird is a type of animal.

Representation

Bird → ISA → Animal

More Examples

Subclass	ISA	Superclass
Bird	ISA	Animal
Dog	ISA	Mammal
Mammal	ISA	Animal
Car	ISA	Vehicle

This means the subclass inherits properties from the superclass.

3. ISA Hierarchy (In Depth)

Meaning: An **ISA hierarchy** is a **multi-level classification structure** where classes are arranged from general to specific.

This structure organizes knowledge **from general categories to specialized objects**.

Example of ISA Hierarchy

Living Thing

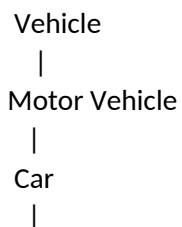


Here:

- Sparrow is a Bird
- Bird is an Animal
- Animal is a Living Thing

Each level represents a **more specific concept**.

Another Example



Electric Car

Each lower level represents a **more specialized type**.

Advantages of ISA Hierarchy

Advantage	Explanation
Organized knowledge	clear structure
Avoids redundancy	shared properties
Supports inheritance	properties flow down

4. Inheritance

Meaning : Inheritance means that a **subclass automatically receives the properties of its parent class**.

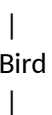
This allows knowledge to be represented efficiently.

Instead of repeating properties for every class, they are **stored at higher levels in the hierarchy**.

Example

Hierarchy:

Animal



Sparrow

Properties:

Class	Property
Animal	breathes
Bird	has wings
Sparrow	small size

Because of inheritance:

Sparrow automatically has:

- breathes
- has wings
- small size

Representation

Animal → breathes

Bird → has wings

Sparrow → small

Properties flow down the hierarchy.

Example of Inheritance in Semantic Network

Animal → breathes

|

ISA

|

Bird → has wings

|

ISA

|

Sparrow

Sparrow inherits properties from Bird and Animal.

Exception Problem in Inheritance

Sometimes inheritance does not apply perfectly.

Example: Bird → can fly

But:

Penguin → Bird

Penguin → cannot fly

This is called **exception handling in inheritance**.

Representing Relationships Between Objects

AI systems must represent relationships between objects in order to understand the world.

Common relationships include:

Relationship	Meaning
instance-of	object belongs to class
ISA	subclass belongs to superclass
part-of	object is part of another
has-property	object has attribute

Example Semantic Network

Sparrow → instance-of → Bird

Bird → ISA → Animal

Animal → has → Life

Bird → has → Wings

From this network the system understands:

- Sparrow is a Bird
- Bird is an Animal

- Sparrow has Wings

Real AI Applications

Application	Example
Knowledge graphs	Google knowledge graph
Ontologies	semantic web
Expert systems	medical classification
Natural language processing	concept hierarchy

Comparison: Instance vs ISA Relationship

Feature	Instance Relationship	ISA Relationship
Type	object → class	class → superclass
Purpose	identify object membership	classify categories
Example	Tom instance-of Cat	Cat ISA Animal

Summary

Concept	Meaning
Instance Relationship	object belongs to class
ISA Relationship	subclass belongs to superclass
ISA Hierarchy	multi-level classification
Inheritance	properties passed from parent to child

These relationships allow AI systems to **organize knowledge hierarchically and perform reasoning efficiently**.

3.5 Computable Functions and Predicates

This topic explains how **logical relationships and functions** are used to represent knowledge in **predicate logic**.

3.5 Computable Functions and Predicates:

In Artificial Intelligence, **predicate logic** is used to represent complex relationships between objects. Unlike propositional logic (which represents simple facts), predicate logic allows us to represent **objects, properties, and relationships**.

Two important concepts in predicate logic are:

- **Functions**
- **Predicates**

These help AI systems represent **structured knowledge and logical relationships between objects**.

Overview

Concept	Meaning
Predicate	describes a property or relationship
Function	maps objects to another object
Logical Expression	combination of predicates and functions
Predicate Logic	logical system for representing knowledge

1. Predicates

Meaning: A **predicate** represents a **property of an object or a relationship between objects**.

It usually has the form: Predicate(Object1, Object2, ...)

The predicate becomes **true or false depending on the objects involved**.

Example

Predicate: Likes(John, IceCream)

Meaning: John likes ice cream.

Here:

Part	Meaning
Likes	predicate
John	object
IceCream	object

More Examples

Predicate Expression	Meaning
Student(Rahul)	Rahul is a student
Father(Ram, Shyam)	Ram is father of Shyam
Greater(5,3)	5 is greater than 3

Predicates describe **relationships or properties**.

Predicate Structure

General form:

Predicate(term1, term2, ...)

Terms can be:

- constants
- variables
- functions

Types of Predicates

Type	Example	Meaning
Unary predicate	Human(John)	property of one object
Binary predicate	Loves(Ram, Sita)	relationship between two objects
Ternary predicate	Between(A,B,C)	relationship among three objects

2. Functions in Logic

Meaning: A **function** in logic represents a **mapping from objects to another object**.

A function takes **input values** and **produces a single output value**.

Function Representation

General form: Function(argument1, argument2)

The function returns a value.

Example

Function: Father(John)

Output: David

Meaning: David is the father of John.

More Examples

Function	Meaning
Age(John)	John's age
Capital(India)	capital of India
Square(4)	square of 4

Example in AI

Consider the statement:

The father of John is David.

Logical representation:

Father(John) = David

Here:

- **Father** is a function
- **John** is the input
- **David** is the output

Difference Between Predicates and Functions

Feature	Predicate	Function
Output	True or False	Object or value
Purpose	describe relationships	produce values
Example	Loves(Ram,Sita)	Father(Ram)

3. Logical Expressions

Meaning : A **logical expression** is formed by combining predicates and functions using logical operators.

Logical expressions represent **complex knowledge structures**.

Example

Knowledge:

1. All humans are mortal
2. Socrates is a human

Logical expression:

Human(x) $\rightarrow$ Mortal(x)

Human(Socrates)

Inference: Mortal(Socrates)

Logical Operators Used

Operator	Symbol	Meaning
AND	$\wedge$	both conditions true
OR	$\vee$	at least one true
NOT	$\neg$	negation
IMPLIES	$\rightarrow$	implication
FOR ALL	$\forall$	universal quantifier
EXISTS	$\exists$	existential quantifier

Example Using Quantifiers

Statement: All humans are mortal.

Representation: $\forall x$ (Human(x) $\rightarrow$ Mortal(x))

Meaning: For every x, if x is human then x is mortal.

Example:

There exists a student who passed the exam.

$\exists x$ (Student(x) $\wedge$ Passed(x))

Meaning: There exists some x such that x is a student and passed.

4. Representing Relationships Between Objects

Predicate logic is powerful because it can represent **complex relationships between objects**.

Example 1

Statement: Ram is the father of Shyam.

Representation: Father(Ram, Shyam)

Example 2

Statement: All birds can fly.

Representation: $\forall x (\text{Bird}(x) \rightarrow \text{CanFly}(x))$

Example 3

Statement: Some students like mathematics.

Representation: $\exists x (\text{Student}(x) \wedge \text{Likes}(x, \text{Mathematics}))$

5. Basic Terms Related to Functions

Several basic terms are used when discussing functions in predicate logic.

Constant : A **constant** represents a specific object.

Examples:

John

India

5

These values do not change.

Variable : A **variable** represents an unspecified object.

Examples:

x

y

z

Variables can represent different values.

Term : A **term** is an element used in predicates.

Terms can be:

- constants
- variables
- functions

Example: Father(John)

This entire expression is a term.

Argument : Arguments are the **inputs to functions or predicates**.

Example: Loves(Ram, Sita)

Arguments:

- Ram
- Sita

Example: Combined Representation

Statement: John loves the daughter of Mary.

Logical representation: Loves(John, Daughter(Mary))

Here:

Part	Type
Loves	predicate
Daughter	function
John	constant
Mary	constant

Importance of Functions and Predicates in AI

Functions and predicates allow AI systems to:

- represent complex relationships
- express general knowledge
- perform logical reasoning
- derive new conclusions

They are widely used in:

AI Application	Example
Expert systems	medical diagnosis
Knowledge graphs	semantic web
Natural language understanding	relation extraction
Automated reasoning	theorem proving

Summary

Concept	Meaning
Predicate	describes property or relationship
Function	maps input objects to output object
Logical Expression	combination of predicates and functions
Constant	fixed object
Variable	unknown object

Functions and predicates make knowledge representation **more powerful and expressive**, allowing AI systems to represent **complex real-world relationships**.

Resolution: This topic explains a **logical inference method** used in AI to **prove conclusions and perform automated reasoning**.

Resolution: Resolution is an important **logical inference technique** used in Artificial Intelligence to derive conclusions from known facts and rules. It is widely used in **automated reasoning and theorem proving systems**.

The main idea of resolution is to **combine logical statements in order to eliminate contradictions and prove that a conclusion is true**.

Resolution is mainly used with **predicate logic and propositional logic**.

Overview of Resolution

Concept	Meaning
Resolution Rule	method used to combine clauses
Logical Inference	deriving new knowledge from known facts
Theorem Proving	proving a statement logically
Resolution Method	algorithm used for automated reasoning

1. Logical Inference

Meaning : Logical inference is the process of **deriving new conclusions from existing knowledge**.

AI systems use inference to:

- discover new information
- verify logical statements
- solve reasoning problems

Example

Given facts:

All humans are mortal

Socrates is a human

Inference: Socrates is mortal

The system derives the conclusion using logical reasoning.

Types of Logical Inference

Method	Description
Deduction	deriving specific conclusions from general rules
Induction	deriving general rules from observations
Abduction	finding best explanation

Resolution is mainly used for **deductive reasoning**.

2. Resolution Rule

Meaning : The **resolution rule** is a logical rule used to **derive new clauses by eliminating complementary literals**.

It combines two logical statements that contain **opposite literals**.

Literal: A **literal** is either:

- a proposition (P)
- a negated proposition ($\neg P$)

Resolution Principle

Example clauses:

$P \vee Q$

$\neg P \vee R$

Since **P** and **$\neg P$** are **complementary**, they cancel each other.

Result: $Q \vee R$

This new clause is called the **resolvent**.

Resolution Formula

General form:

$(A \vee B)$

$(\neg B \vee C)$

Resolution result: $A \vee C$

The complementary literal **B** and **$\neg B$** are eliminated.

Example of Resolution

Given clauses:

1. $P \vee Q$

2. $\neg P$

Apply resolution: Q

Reason:

- P and $\neg P$ cancel each other

- remaining clause is Q

3. Resolution in Propositional Logic

Resolution can be applied to propositional logic by converting statements into **clause form**.

Steps in Resolution

1. Convert statements to propositional logic
2. Convert to **Conjunctive Normal Form (CNF)**
3. Apply resolution rule
4. Continue until contradiction or conclusion is obtained

Example

Statements:

1. If it rains, the ground is wet
2. It is raining

Representation:

Rain $\rightarrow$ Wet

Rain

Convert implication:

$\neg \text{Rain} \vee \text{Wet}$

Rain

Resolution result: Wet

Conclusion: the ground is wet.

4. Resolution in Predicate Logic

In predicate logic, resolution also involves **unification**.

Unification means matching variables with constants.

Example

Statements:

$\text{Human}(x) \rightarrow \text{Mortal}(x)$

$\text{Human}(\text{Socrates})$

Convert implication:

$\neg \text{Human}(x) \vee \text{Mortal}(x)$

$\text{Human}(\text{Socrates})$

Unification: $x = \text{Socrates}$

Resolution result: $\text{Mortal}(\text{Socrates})$

5. Theorem Proving

Meaning : Theorem proving is the process of **proving that a statement logically follows from a set of known facts and rules**.

Resolution is widely used in automated theorem proving.

Example

Facts:

All birds have wings

Sparrow is a bird

Representation:

Bird(x) $\rightarrow$ HasWings(x)
Bird(Sparrow)
Resolution result: HasWings(Sparrow)
The system proves the theorem logically.

Refutation Method

Resolution often uses a technique called **refutation**.

Steps:

1. Assume the **negation of the conclusion**
2. Apply resolution with known facts
3. If contradiction appears $\rightarrow$ conclusion is true

Example

Goal: Prove Q

Steps:

1. Add $\neg Q$
2. Apply resolution
3. If contradiction occurs $\rightarrow$ Q is true

Resolution Process

General algorithm:

1. Convert all statements to CNF
2. Negate the goal statement
3. Add it to the knowledge base
4. Apply resolution rule repeatedly
5. If empty clause is obtained $\rightarrow$ theorem proven

The empty clause represents **contradiction**.

Advantages of Resolution

Advantage	Explanation
powerful inference method	derives new knowledge
systematic reasoning	structured problem solving
suitable for automation	used in AI theorem provers

Limitations

Limitation	Explanation
computationally expensive	many possible clauses
requires CNF conversion	extra processing
may generate many intermediate clauses	

Applications of Resolution in AI

Application	Example
Automated theorem proving	mathematical reasoning
Logic programming	Prolog systems
Expert systems	rule-based inference
Knowledge-based systems	logical reasoning

Example of Resolution Tree

$P \vee Q$

$\neg P \vee R$

$\neg R$

Resolution steps:

1.

$P \vee Q$

$\neg P \vee R$

Result:

$Q \vee R$

2.

$Q \vee R$

$\neg R$

Result: Q

Final conclusion: **Q is true.**

Summary

Concept	Meaning
Resolution	logical rule for deriving conclusions
Logical Inference	deriving new knowledge from facts
Resolution Rule	eliminates complementary literals
Theorem Proving	proving statements logically
Refutation	proving by contradiction

Resolution is one of the **most important reasoning techniques in AI**, enabling systems to **perform automated logical reasoning and prove conclusions from knowledge bases**.

3.7 Procedural vs Declarative Knowledge: This topic explains **two fundamental ways knowledge is stored in AI systems**.

Procedural vs Declarative Knowledge : In Artificial Intelligence, knowledge can be stored in different forms depending on how it is used. Two of the most important forms are:

1. **Procedural Knowledge**
2. **Declarative Knowledge**

These represent **two major ways of storing and organizing knowledge in AI systems**.

- **Declarative knowledge** describes *what is known*.
- **Procedural knowledge** describes *how to perform tasks*.

Overview

Knowledge Type	Meaning	Focus
Declarative Knowledge	knowledge of facts and information	WHAT
Procedural Knowledge	knowledge of procedures or steps	HOW

1. Declarative Knowledge (In Depth)

Meaning : **Declarative knowledge** represents **facts and information about the world**. It describes knowledge in a way that states **what is true** without specifying how to use it.

This knowledge is usually expressed as:

- logical statements
- facts
- relationships between objects

Declarative knowledge is also called **descriptive knowledge**.

Example

Facts about geography:

Capital(France, Paris)

Capital(India, NewDelhi)

Facts about people:

Student(Rahul)

Age(Rahul, 20)

These statements describe **facts about the world**.

Representation of Declarative Knowledge

Declarative knowledge is commonly represented using:

Representation Method	Example
Predicate logic	Human(x) $\rightarrow$ Mortal(x)
Semantic networks	Dog $\rightarrow$ ISA $\rightarrow$ Animal
Frames	object attributes
Databases	relational tables

Example in AI

Knowledge base:

Human(x) $\rightarrow$ Mortal(x)

Human(Socrates)

Inference: Mortal(Socrates)

The system stores facts and derives conclusions.

Characteristics of Declarative Knowledge

Characteristic	Explanation
Explicit representation	facts clearly stored
Easy to modify	new facts can be added
Supports reasoning	inference can derive new knowledge
Independent of procedure	does not specify how knowledge is used

Advantages of Declarative Knowledge

Advantage	Explanation
easy to understand	similar to human knowledge
flexible	knowledge can be updated easily
supports logical inference	new facts can be derived

Limitations

Limitation	Explanation
slower execution	reasoning required
large knowledge bases	may become complex

2. Procedural Knowledge (In Depth)

Meaning: Procedural knowledge represents knowledge as **procedures or sequences of steps required to perform tasks**.

It focuses on **how something is done** rather than describing facts.

Procedural knowledge is often implemented as:

- algorithms
- programs
- procedures

Example

Procedure for calculating factorial:

factorial(n):

if $n == 1$ return 1

else return $n \times \text{factorial}(n-1)$

This represents **how to compute factorial**.

Example: Making Tea

Steps:

1. Boil water
2. Add tea leaves
3. Add sugar
4. Add milk

This describes **how to prepare tea**.

Representation of Procedural Knowledge

Procedural knowledge is usually stored in:

Representation	Example
Programs	computer algorithms
Production rules	IF-THEN rules
Scripts	task sequences
Planning procedures	action sequences

Example in AI

Rule-based system:

IF patient has fever

THEN check infection

This rule specifies **what action should be performed**.

Characteristics of Procedural Knowledge

Characteristic	Explanation
action-oriented	describes procedures
efficient execution	direct instructions
task-focused	used for problem solving
embedded knowledge	knowledge inside procedures

Advantages of Procedural Knowledge

Advantage	Explanation
fast execution	procedures executed directly
useful for task automation	ideal for algorithms
efficient for problem solving	reduces reasoning overhead

Limitations

Limitation	Explanation
------------	-------------

difficult to modify	changing procedure may affect system
knowledge not explicit	harder to understand
limited reasoning	facts not easily derived

Example Comparing Both Knowledge Types

Consider cooking rice.

Declarative knowledge:

Rice is a grain

Rice requires water to cook

Rice cooks at high temperature

Procedural knowledge:

1. Wash rice

2. Add water

3. Heat until cooked

Both types represent the **same domain knowledge differently**.

Differences Between Procedural and Declarative Knowledge

Feature	Declarative Knowledge	Procedural Knowledge
Focus	what is known	how to perform tasks
Representation	facts and logical statements	procedures and algorithms
Execution	requires reasoning	direct execution
Flexibility	easier to modify	difficult to modify
Example	"Paris is capital of France"	steps to solve equation

Relationship Between Both Types

Many AI systems use **both types of knowledge together**.

Example: Medical expert system

Knowledge Type	Example
Declarative knowledge	symptoms of diseases
Procedural knowledge	diagnosis procedure

Combining both allows AI systems to **store knowledge and apply it effectively**.

Real-World Applications

Application	Declarative Knowledge	Procedural Knowledge
Expert systems	knowledge base	inference rules
Robotics	environment facts	motion planning
Databases	stored information	query execution

Importance in AI

Understanding these knowledge types helps AI designers:

- choose proper representation
- design efficient reasoning systems
- organize knowledge effectively

Most modern AI systems use a **hybrid approach combining both forms**.

Summary

Concept	Meaning
Declarative Knowledge	describes facts about the world

Procedural Knowledge	describes procedures to perform tasks
Declarative focus	WHAT
Procedural focus	HOW

Both forms of knowledge representation are essential for building **intelligent AI systems capable of reasoning and performing tasks**.

3.8 Logic Programming

This topic explains **how programming languages like Prolog use logical rules to solve problems automatically**.

Logic Programming : Logic Programming is a programming paradigm where programs are written using **logical statements, facts, and rules**, and the system automatically derives solutions using logical inference.

Instead of writing detailed step-by-step instructions (like in procedural programming), the programmer only describes **facts and relationships**, and the computer **reasons logically to find answers**.

Logic programming is mainly based on **predicate logic**.

A well-known language for logic programming is **Prolog (Programming in Logic)**.

Overview

Concept	Meaning
Logic Programming	programming based on logical rules
Facts	basic truths about objects
Rules	logical relationships between facts
Queries	questions asked to the system
Inference	process used to derive answers

1. Concept of Logic Programming

Meaning : Logic programming is a programming method where **knowledge is represented as logical facts and rules**, and the system uses **logical reasoning to solve problems**.

Instead of specifying how to solve a problem, the programmer specifies **what is true about the problem domain**.

The inference engine then determines the solution.

Key Idea

Traditional programming:

Step-by-step instructions

Logic programming: Facts + Rules → Logical inference → Solution

Example

Knowledge:

All humans are mortal

Socrates is a human

Logic representation:

Human(Socrates)

Human(x) → Mortal(x)

Query: Is Socrates mortal?

The system infers: Mortal(Socrates)

Structure of Logic Programming System

A logic programming system usually contains three components.

Component	Description
Knowledge Base	collection of facts and rules
Inference Engine	reasoning mechanism
Query System	interface to ask questions

2. Basic Terminology of Logic Programming

Several important terms are used in logic programming.

1. Facts : Facts are **basic statements that are always true**.

They represent known information.

Example:

human(socrates).

student(rahul).

Meaning:

- Socrates is a human
- Rahul is a student

Facts are stored in the **knowledge base**.

2. Rules : Rules describe **logical relationships between facts**.

They usually have the form: Head :- Body.

Meaning: Head is true if Body is true

Example

mortal(X) :- human(X).

Meaning: If X is human, then X is mortal.

3. Query : A **query** is a question asked to the system.

Example: ?- mortal(socrates).

The system checks facts and rules to determine whether the query is true.

4. Predicate: A **predicate** represents a property or relationship.

Example:

likes(rahul, cricket).

Predicate: **likes**

Arguments:

- rahul
- cricket

5. Clause

A **clause** is a logical statement used in logic programming.

Two main types:

Clause Type	Example
Fact clause	human(socrates).
Rule clause	mortal(X) :- human(X).

6. Atom

An **atom** is a basic logical expression containing:

- predicate

- arguments

Example:

student(rahul)

7. Variable

A variable represents an **unknown value**.

Examples:

X

Y

Z

Example rule: mortal(X) :- human(X).

3. Rule-Based Programs

Meaning : Rule-based programs represent knowledge using **IF-THEN rules**.

Rules describe **how conclusions can be derived from known facts**.

These systems are widely used in **expert systems**.

Structure of Rule-Based System

Rule format:

IF condition

THEN action

Example

Medical knowledge:

IF patient_has_fever

THEN check_for_infection

Logic programming representation:

infection(X) :- fever(X).

Example Rule-Based Program

Facts:

bird(sparrow).

bird(parrot).

Rule: can_fly(X) :- bird(X).

Query: ?- can_fly(sparrow).

Answer: Yes

Example: Family Relationship Program

Facts:

parent(john, mary).

parent(mary, alice).

Rule: grandparent(X,Y) :- parent(X,Z), parent(Z,Y).

Query: ?- grandparent(john, alice).

Answer: True

The system uses logical inference to derive the result.

Logic Programming Execution Process

The system works in the following steps:

Step	Description
------	-------------

Step 1	store facts and rules
Step 2	user submits query
Step 3	inference engine searches knowledge base
Step 4	logical inference applied
Step 5	result returned

Advantages of Logic Programming

Advantage	Explanation
simple knowledge representation	facts and rules
powerful reasoning	automatic inference
easy modification	new rules can be added
good for expert systems	knowledge-based systems

Limitations

Limitation	Explanation
slower execution	reasoning process required
complex rules may be difficult	large knowledge bases
requires logical structure	knowledge must be formalized

Applications of Logic Programming

Application	Example
Expert systems	medical diagnosis
Natural language processing	language understanding
Automated reasoning	theorem proving
Knowledge-based systems	decision support systems

Comparison: Logic Programming vs Procedural Programming

Feature	Logic Programming	Procedural Programming
Focus	what is true	how to perform tasks
Representation	facts and rules	instructions
Execution	inference based	step-by-step
Example language	Prolog	C, Java

Summary

Concept	Meaning
Logic Programming	programming using logical rules
Facts	known truths
Rules	logical relationships
Queries	questions to system
Rule-Based Programs	programs using IF-THEN rules

Logic programming allows AI systems to **solve problems using logical reasoning instead of explicit step-by-step instructions**.

3.9 Forward Reasoning (Forward Chaining) and 3.10 Backward Reasoning (Backward Chaining) — two important inference methods used in AI reasoning systems.

Forward Reasoning (Forward Chaining) : **Forward reasoning**, also called **Forward Chaining**, is an inference technique used in Artificial Intelligence where reasoning starts from **known facts and applies rules to derive new conclusions**.

It is called **data-driven reasoning** because the reasoning process begins with available data and moves forward until a goal or conclusion is reached.

This method is widely used in:

- expert systems
- rule-based systems
- knowledge-based systems

Overview

Concept	Meaning
Forward Reasoning	reasoning that starts from facts
Forward Chaining	inference method used to apply rules
Data-Driven Reasoning	reasoning guided by available data
Inference Engine	component that applies rules

1. Reasoning in AI

Meaning : Reasoning in AI is the process of **deriving new knowledge from existing knowledge using logical rules.**

AI systems use reasoning to:

- solve problems
- make decisions
- answer queries

Example

Knowledge:

All humans are mortal

Socrates is human

AI reasoning: Socrates is mortal

The system derives new information using logical rules.

Types of Reasoning in AI

Type	Description
Forward Reasoning	start from facts and derive conclusions
Backward Reasoning	start from goal and verify facts
Deductive Reasoning	general rule → specific result
Inductive Reasoning	observations → general rule

Forward reasoning mainly uses **deductive reasoning**.

2. Basic Terms Related to Forward Reasoning

Several important components are used in forward reasoning.

Knowledge Base : The **knowledge base** stores all facts and rules used by the system.

Example:

Human(Socrates)

Human(x) → Mortal(x)

Facts

Facts represent known information.

Example: Human(Socrates)

Rules

Rules define logical relationships.

Example: $\text{Mortal}(x) :- \text{Human}(x)$

Meaning: If x is human, then x is mortal.

Inference Engine

The **inference engine** is the component that applies rules to facts in order to derive new knowledge.

Functions of inference engine:

- match rules with facts
- generate new conclusions
- update knowledge base

Working Memory

Working memory contains **current facts used during reasoning**.

As new conclusions are derived, they are added to working memory.

3. Forward Chaining Inference

Meaning : Forward chaining is an inference technique where the system **starts with known facts and repeatedly applies rules to derive new facts** until the goal is reached.

It works in a **step-by-step forward direction**.

Basic Idea

Facts → Apply Rules → Generate New Facts → Continue → Goal Achieved

The system continuously expands knowledge using rules.

Forward Chaining Process

Step	Description
Step 1	Start with known facts
Step 2	Find rules whose conditions match facts
Step 3	Apply rule and generate new fact
Step 4	Add new fact to knowledge base
Step 5	Repeat until goal is reached

Example

Knowledge:

Facts: rain

Rules:

Rain → WetGround

WetGround → SlipperyRoad

Forward Reasoning Steps

Step 1:

Rain

Apply rule: Rain → WetGround

Result: WetGround

Step 2:

Apply rule: WetGround → SlipperyRoad

Result: SlipperyRoad

Final conclusion: SlipperyRoad

Example in Medical Diagnosis

Facts:

Patient has fever

Patient has cough

Rule: IF fever AND cough THEN infection

Forward chaining: fever $\wedge$ cough $\rightarrow$ infection

Conclusion: Patient has infection

Forward Chaining Algorithm

General algorithm:

1. Initialize working memory with known facts
2. Check rules whose conditions match facts
3. Fire the rule and generate new facts
4. Add new facts to working memory
5. Repeat until goal is reached or no rules apply

Data-Driven Reasoning

Forward chaining is called **data-driven reasoning** because reasoning begins with **available data**. The system gradually builds conclusions using data.

Data-Driven Reasoning Flow

Data (Facts)



Rule Matching



Rule Execution



New Facts



Goal

The process continues until no new knowledge can be generated.

Example: Animal Identification

Facts:

Animal has feathers

Animal lays eggs

Rules:

IF feathers THEN bird

IF bird THEN can_fly

Forward reasoning:

Step 1: feathers $\rightarrow$ bird

Step 2: bird $\rightarrow$ can_fly

Conclusion:

Animal can fly

Advantages of Forward Reasoning

Advantage	Explanation
automatic reasoning	system derives new knowledge

useful when many facts available	data-driven systems
good for monitoring systems	real-time data processing

Limitations

Limitation	Explanation
may generate unnecessary conclusions	explores many rules
inefficient for specific goals	may examine many possibilities

Applications of Forward Reasoning

Application	Example
Expert systems	medical diagnosis
Rule-based systems	decision support
Monitoring systems	fault detection
AI planning	automated decision making

Example of Forward Chaining Tree

Facts

|

v

Rule 1 → New Fact

|

v

Rule 2 → New Fact

|

v

Goal Achieved

The reasoning progresses step by step.

Summary

Concept	Meaning
Forward Reasoning	reasoning from facts toward conclusion
Forward Chaining	inference technique applying rules
Data-Driven Reasoning	reasoning guided by data
Knowledge Base	store of facts and rules
Inference Engine	applies rules to derive knowledge

Forward reasoning is an important AI technique that allows systems to **derive new knowledge automatically from available information**.

3.10 Backward Reasoning (Backward Chaining) — a goal-driven inference method used in expert systems and logic programming.

Backward Reasoning (Backward Chaining) : Backward reasoning, also called **Backward Chaining**, is an inference technique used in Artificial Intelligence where reasoning **starts from the goal (desired conclusion) and works backward to check whether the known facts support that goal**.

It is called **goal-driven reasoning** because the system begins with a goal and searches for facts or rules that prove the goal is true.

Backward reasoning is widely used in:

- expert systems
- logic programming (Prolog)

- diagnostic systems

Overview

Concept	Meaning
Backward Reasoning	reasoning starting from goal
Backward Chaining	inference method working backward
Goal-Driven Reasoning	reasoning guided by a target goal
Inference Engine	component that performs reasoning
Knowledge Base	collection of facts and rules

1. Backward Reasoning

Meaning : Backward reasoning begins with a **goal (hypothesis)** and tries to determine whether it is true by examining available facts and rules.

The system asks:

- What must be true for this goal to be true?

Then it checks whether those conditions are satisfied.

Basic Idea

Goal → Find Rule → Check Conditions → Verify Facts → Goal Proven

The reasoning process continues until:

- the goal is proven, or
- no supporting facts are found.

2. Backward Chaining

Meaning : **Backward chaining** is the inference technique used to implement backward reasoning.

The system works **from goal to facts**, repeatedly breaking the goal into **subgoals**.

Backward Chaining Process

Step	Description
Step 1	Start with a goal
Step 2	Find rules that produce the goal
Step 3	Identify rule conditions
Step 4	Check whether conditions are true
Step 5	If conditions are true → goal proven

Example

Knowledge:

Rules:

Rain → WetGround

WetGround → SlipperyRoad

Facts: Rain

Goal: SlipperyRoad

Backward Reasoning Steps

Step 1:

Goal: SlipperyRoad

Look for rule: WetGround → SlipperyRoad

Subgoal: WetGround

Step 2:

Check rule:

Rain $\rightarrow$ WetGround

Subgoal:

Rain

Step 3:

Fact exists:

Rain

Therefore:

WetGround

SlipperyRoad

Goal proven.

3. Knowledge Base

Meaning : The **knowledge base** is the storage area containing **facts and rules used for reasoning**. The reasoning system searches the knowledge base to determine whether the goal can be proven.

Components of Knowledge Base

Component	Description
Facts	known information
Rules	logical relationships

Example

Facts: human(socrates)

Rule: mortal(X) :- human(X)

Goal: mortal(socrates)

The system searches the knowledge base to verify the goal.

4. Inference

Meaning : Inference is the process of **deriving new knowledge from existing facts and rules**.

Inference allows AI systems to:

- draw conclusions
- answer queries
- solve problems

Backward reasoning is one method used for inference.

Types of Inference

Type	Description
Forward chaining	facts $\rightarrow$ goal
Backward chaining	goal $\rightarrow$ facts

5. Inference Engine

Meaning: The **inference engine** is the reasoning component of an AI system. It applies logical rules from the knowledge base to derive conclusions.

Functions of Inference Engine

Function	Description
rule matching	identify applicable rules
goal evaluation	determine whether goal is satisfied
reasoning control	manage inference process

6. Goal-Driven Reasoning

Backward reasoning is called **goal-driven reasoning** because it begins with a goal.

The system asks:

What conditions must be true for this goal?

Then it checks the knowledge base to verify those conditions.

Example: Medical Diagnosis

Goal: Patient has infection

Rule:

IF fever AND cough → infection

Backward reasoning:

Check:

Does patient have fever?

Does patient have cough?

If both facts are true, the goal is proven.

Backward Chaining Algorithm

General algorithm:

1. Start with goal
2. Search rules that conclude the goal
3. Identify rule conditions
4. Check if conditions are facts
5. If not, create subgoals
6. Repeat until goal proven

Example: Animal Identification

Goal: Animal is bird

Rule:

IF feathers AND lays eggs → bird

Backward reasoning:

Check:

Does animal have feathers?

Does animal lay eggs?

If both are true → conclusion confirmed.

Difference Between Forward and Backward Reasoning

Feature	Forward Reasoning	Backward Reasoning
Starting point	facts	goal
Reasoning direction	data → conclusion	goal → facts
Approach	data-driven	goal-driven
Efficiency	good when many facts available	good when specific goal exists
Example	monitoring systems	diagnostic systems

Example Comparison

Forward reasoning: Facts → Rules → New Facts → Goal

Backward reasoning: Goal → Rules → Subgoals → Facts

Applications of Backward Reasoning

Application	Example
Expert systems	medical diagnosis
Logic programming	Prolog
Fault diagnosis	troubleshooting systems
Question answering	AI assistants

Summary

Concept	Meaning
Backward Reasoning	reasoning starting from goal
Backward Chaining	inference technique working backward
Goal-Driven Reasoning	goal guides reasoning process
Knowledge Base	store of facts and rules
Inference Engine	component performing reasoning

Backward reasoning allows AI systems to **verify goals efficiently by checking whether supporting facts and rules exist in the knowledge base.**